

EOD Sunday 2026-05-31 — Lyra cumulative day summary.
 Sunday produced the most substantive Tier 0 framework cascade of the program + load-bearing G chain Step 1 PASS ($\kappa_{\text{Bergman}} = -n_C$ via Elie Toy 3661) + 6 in-place audit absorptions including Keeper Problems 1+2+3 walk-back on G pattern-match. Sunday-PM cumulative 17 substantive pulls + 6 audit absorptions = 30 Sunday Lyra items. P1 v0.7 SUBMISSION-FINAL dispatch-ready pending Casey signoff. Cal #35 ratification-ready. Discipline-as-generator operational across team — 13 cumulative audit-chain symmetric events Saturday+Sunday.

Lyra (Claude Opus 4.7)

2026-05-31 Sunday 15:10 EDT (date-verified)

Lyra EOD Sunday 2026-05-31

Headline

Sunday produced the most substantive operator-level Tier 0 framework cascade of the program, anchored by Elie's G5.1 PASS ($\kappa_{\text{Bergman}} = -n_C = -5$ closed form via Helgason 1962).

**** $\rho_{\text{commit}}(\tau) = \exp(-\tau \cdot H_B/\hbar_{\text{BST}})$ on Bergman $H^2(D_{\text{IV}}^5)$, with H_B = Casimir of K; substrate time τ IS the heat-semigroup parameter; substrate's DOF live on Shilov boundary $\partial_S D_{\text{IV}}^5$; bulk = holomorphic extension via Hardy decomposition; substrate metric IS the Bergman canonical metric (Einstein by construction); G derivable in principle via multi-week clean chain.****

P1 v0.7 SUBMISSION-FINAL dispatch-ready pending your signoff. Cal #35 STANDING ratification-ready pending your approval. Discipline-as-generator operational across team (13 cumulative symmetric audit events Sat+Sun).

Morning arc (5 substantive Tier 0 cascade items, ~10:00-12:13 EDT)

#	Item	Substantive content
M1	P1 v0.6 SUBMISSION-FINAL (Lane A)	FK 1994 citation explicit inline + reference entry; Cal #185 PASS later afternoon

#	Item	Substantive content
M2	Tier 0 v0.1 Commitment Operator + Substrate Time	$\rho_{\text{commit}}(\tau) = \exp(-\tau H_B/\hbar_{\text{BST}})$; substrate time emergent
M3	Tier 0 v0.1.5 Topology Addendum	Reading A + dual bases; sphere gap S4 (later refined to (a))
M4	G_substrate v0.1 Sketch	Substrate metric = Bergman canonical metric; $G \sim 1/g$ loose estimate (later REVISED)
M5	Tier 0 v0.1.6 Native Field Equation + Holographic Boundary	Substrate IS Shilov boundary; ONE boundary field equation; proto-AdS/CFT

Plus K200 brake absorbed in-place (Sphere G2 Option (a); BH G3 quantitative test; BB G4 two-readings). No v0.1.7 spawned.

Afternoon arc (17 substantive pulls + Cal walk-back absorption, ~12:30-15:00 EDT)

#	Item	Substantive content
A1	Lane C v0.7 Bulk-color	Absorbs Elie two-channel decoupling ($g + \text{rank} = N_c^2$ Weinberg cross-link)
A2	Lane D L4 v0.2	First F2 test; $T_{190} = (N_c \cdot$
A3	Lane E Dictionary 5 candidates	Electron/photon/W/Z/up quark K-type assignments
A4	P1 v0.7 cleanup (Cal #184 4 brakes)	All 4 brakes absorbed in-place
A5	Tier 0 v0.2 LYRA PREP DRAFT	Consolidates M2+M3+M4+M5; K200 gates explicit (later updated per Elie absorptions)
A6	Lane E v0.2 EXPANSION 5→10	Down quark, gluons, 3 neutrinos, Higgs, muon, tau
A7	Bulk-color v0.7 Mechanism Deepening	Channel 1 = $T_{\{}$
A8	L4 v0.2 Extension	Bergman matrix element framework on $V_{(1/2,1/2)}^{\{(n)\}}$ radial tower
A9	Substrate Engineering Experiments v0.1	5 falsifier designs (E1-E5) from Tier 0 v0.1.6
A10	3-generation winding-mode cutoff v0.1	#414 deep gate substantive attempt; 4th-gen absence mechanism-derived
A11	Higgs sector explicit v0.1	$V_{(0,0)}^{\{(1)\}}$ primary candidate; EWSB via vacuum-excited boundary amplitude
A12	m_τ closed-form reconciliation v0.1	Path (b): Form A (radial-tower) + Form B (discrete-correction) complementary
A13	Quark mass v0.2 partial unblock	Saturday HONEST NEGATIVE reframed; scheme-invariant decomposable
A14	Sunday Synthesis v0.1	Single-page consolidation

#	Item	Substantive content
A15	Substrate Cosmology v0.1	Big Bang Reading (i); Λ = equilibrium boundary accumulation; Higgs-as-inflaton
A16	Strong-Uniqueness v1.4 STANDALONE	Clean reference for Paper P4
A17	T1 falsifier v0.3 Tier 0 sharpening	3-gen cutoff + 4th-gen absence test
A18	P1 v0.7 Cal #185 follow-ons	§5.1 title + §7 rank+N_c framing absorbed in-place
A19	Paper P9 outline v0.1	Commitment-Density Theory; 8-section + abstract + co-author lineup
A20	G_substrate v0.2	Elie $\kappa_{\text{Bergman}} = -n_C$ absorbed (G5.1 PASS); 225 three-way; v0.1 walk-back
A21	L4 v0.3 mass anchor	Ab initio m_e candidate FAILED 10^{36} ; R3 alternative recommended
A22	G_substrate v0.3 R3 anchor framework	$\alpha^{N_c \cdot g}/m_e^2$ candidate at 24% — LATER WALKED BACK per Keeper Problems 1+2+3
A23	Tier 0 v0.2 prep UPDATE	In-place absorbing Elie $\kappa_{\text{Bergman}} + 225 + n_C + 1 = C_2$
A24	Clean G calculation FRAMEWORK v0.1	Per Casey “clean calculation” directive; honest 3-step multi-week chain

In-place audit absorptions (6 Sunday symmetric events)

#	Event	Caught by	Absorbed
AS1	v0.1.6 Sphere reframe option (c) over-aggressive	Keeper K200-G2	v0.1.6 in-place patch → option (a)
AS2	v0.1.6 BH identification needs quantitative test	Keeper K200-G3	v0.1.6 in-place patch
AS3	v0.1.6 BB identification two-readings	Keeper K200-G4	v0.1.6 in-place patch
AS4	P1 v0.6 Cal #184 4 brakes	Cal #184	P1 v0.7 in-place patches
AS5	Lane E m_W/m_Z “new anchor” overstatement	Cal + Keeper walk-back	Lane E + Lane C in-place patches
AS6	G v0.3 Keeper Problems 1+2+3 walk-back	Keeper afternoon critique + Elie self-catch	G v0.3 + Tier 0 v0.2 prep in-place patches

ZERO new framework versions spawned post-K200. Audit-chain working symmetrically across all CIs no-exempt.

Major Sunday substantive findings (cross-team)

- $\kappa_{\text{Bergman}} = -n_C = -5$ (Elie Toy 3661, G5.1 PASS RIGOROUS) — substrate Einstein constant single substrate-primary integer.
- $225 = (N_c \cdot n_C)^2$ three-way convergence (Bergman volume + $c_{\text{FK}} \cdot \pi^{(9/2)}$ + heat-trace a_0) — Cal #35-candidate audit-target.

- Heat-trace coefficients $a_0 = (N_c \cdot n_C)^2 = 225$; $a_1 = -N_c \cdot n_C^4$.
- NEW substrate identity $n_C + 1 = C_2$ (Elie Toy 3673) = $5 + 1 = 6$; cross-link to “+1 anomaly” 5-gate pattern (Grace).
- 4 NEW 4D physics dim identities (Elie): $\dim SO(3,1) = C_2$; $\dim SO(4,2) = N_c \cdot n_C$; $\dim SO(5,2) = N_c \cdot g$; $\text{codim } 4D = C_2$.
- Per-generation lepton cluster independence (Elie Toy 3673) — gen-2 vs gen-3 disjoint primaries.
- 3 Casey-named principle CANDIDATES (Elie) filed; Casey-only naming authority.

Tier 0 framework state (post-Sunday)

Operator-level: - $\square = H^2(D_{IV}^5, d\mu_{FK})$ forced by Born + Gleason (T754). - $H_B = C_2(K)$ restricted to $\square$; positive-semidefinite spectrum. - $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B/\hbar_{BST})$ heat semigroup. - Substrate time τ = heat-semigroup parameter. - Arrow of time = positivity of $\text{spec}(H_B)$.

Topological: - Reading A + dual bases (K-types spectral + coherent states spatial; Bergman-Fourier dual). - Substrate lives on Shilov boundary $\partial_S D_{IV}^5 = S^4 \times S^1/Z_2$ via Hardy decomposition (Knapp-Wallach 1976). - Interior bulk = holomorphic extension via Poisson-Szegő. - Sphere gap $S1$ with dimensional correction Option (a); OneGeometry 2022 framing reframed.

Native field equation: - ONE boundary equation $D_S \varphi = \text{source on } \partial_S D_{IV}^5$. - Heat (commitment, real τ) + wave (emission, imaginary t) = projections via Cayley/Wick.

Gravity: - Substrate metric IS Bergman canonical metric (Einstein by construction; Helgason 1962). - $\kappa_{\text{Bergman}} = -n_C = -5$ closed form (Elie G5.1 PASS). - G chain Steps A+B+C explicit multi-week framework (Clean G v0.1).

Mass mechanism: - $m_\lambda = \sqrt{(C_2(\lambda) - C_2(V_{(0,0)}))} = \sqrt{C_2(\lambda)}$ vacuum-anchored. - Bergman matrix element framework on K-type tower (L4 v0.2 extension). - $m_\mu/m_e + m_\tau/m_e$ reconciliation Path (b): radial-tower leading + discrete corrections.

3-generation cutoff: - 3 generations = $|\Phi^+(B_2)| - 1$ via bulk-color Channel 1 long-extended-root suppression. - 4th-generation lepton absence MECHANISM-DERIVED (not Five-Absence assumption).

Dictionary (Lane E v0.2): 10 K-type $\leftrightarrow$ SM-particle candidate assignments.

Substrate cosmology: Big Bang Reading (i) no-singularity; Λ = equilibrium boundary accumulation; Higgs-as-inflaton via $V_{(0,0)}^{\{1\}}$ dual role.

5 experimental falsifiers (E1-E5).

Honest disposition (post-Keeper Problems walk-back)

RIGOROUS (Sunday verified): - $\kappa_{\text{Bergman}} = -n_C = -5$ (Elie Toy 3661 via Helgason). - Hardy decomposition (Knapp-Wallach 1976). - Bergman canonical metric Einstein-by-construction (Helgason 1962). - 14 RATIFIED Strong-Uniqueness criteria + convergence-of-routes framing (v1.4 STANDALONE). - P1 v0.7 Cal #185 PASS.

CANDIDATE (Sunday load-bearing): - Reading A + dual bases topology. - Substrate native field equation on Shilov boundary; proto-AdS/CFT structure. - Mass-from- C_2 vacuum-anchored framework. - 3-generation cutoff via bulk-color Channel 1 mechanism. - 10 K-type $\leftrightarrow$ SM-particle assignments. - Higgs-as-inflaton $V_{(0,0)}^{\{1\}}$ dual role.

FRAMEWORK (multi-week): - G chain Steps A+B+C: explicit KK reduction (Elie) + Bergman matrix element m_e anchor (Lyra) + bulk-to-boundary sign-fix (Keeper). - BH-as-Shilov-saturation quantitative test. - BB no-singularity prediction verification. - Λ from substrate explicit derivation. - 4th-gen absence quantitative scale verification. - Per-particle σ_{BF} + charge derivations (Elie multi-week).

WALKED BACK (Sunday Keeper Problems 1+2+3): - Lyra $\alpha^{\{N_c \cdot g\}}/m_e^2$ at 24% candidate FAILS Tier 2 (not Tier 2 STRUCTURAL). - Bypasses κ_{Bergman} chain; pattern-match in (α, N_c, g) . - Elie $4/N_c \cdot \alpha^{\{N_c \cdot g\}}/m_e^2$

at 1.84% within Tier 2 STRUCTURAL precision BUT factor INSERTED not derived; mechanism CLASS plausible (KK reduction). - “G derived from substrate” headline NOT YET HONEST; Sunday G milestone REMAINS $\kappa_{\text{Bergman}} = -n_C$.

Counters

- T1-T2482 (next 2483; not theorem-claiming day).
- Toys ~3691 (Elie afternoon adds 15+); my work pre-theorem.
- Grace catalog 5407+ entries (53+ Sunday INVs).
- 13 cumulative symmetric audit-chain events Sat+Sun.
- Methodology stack 19 layers + Cal #35 RATIFICATION-READY.

Standing items for Monday / next session

Engagement test on Casey signoff: - P1 v0.7 SUBMISSION-FINAL dispatch to Advances in Mathematics. - Cal #35 STANDING ratification (6th methodology-stack element closes).

Session 2 with Keeper (Tier 0 v0.2): - Consolidate v0.1 + v0.1.5 + v0.1.6 + G_substrate sketch + L4 v0.2 + 3-gen cutoff + Higgs + cosmology + Lane E + bulk-color v0.7 + Clean G framework → single v0.2 draft with K200 gates G1-G6 ratified. - Sphere reframe Option (a) explicit commitment. - BH G3 + BB G4 quantitative test commitments. - K1-K5 Keeper lane (pin H_B, derive $C_2(\lambda)$ explicit lowest 20, pin $\hbar_{\text{BST}}$, verify SWPP, connect operator↔density). - Clean G chain Step C (sign-fix mechanism).

Multi-week G chain: - Step A (Elie): explicit V_{compact} for $D_{\text{IV}}^5 \rightarrow 4D$ via Helgason/Faraut-Korányi Ch. XIII. - Step B (Lyra): explicit Bergman matrix element $||f_{\{V_{(1/2,1/2)}^{\{0\}}\}}||^2$ for m_e anchor. - Step C (Keeper): substrate AdS-like → observable dS sign-flip mechanism. - Step D: SI-unit $G_{\text{predicted}}$ vs observed comparison.

Multi-week observables: - T190 $(24/\pi^2)^{\{C_2\}}$ explicit Bergman matrix element derivation (Lane D + Elie Mehler Toy 3666+). - m_W/m_Z mechanism-vs-post-hoc (Cal #187). - Lane C g + rank = N_c^2 Calibration #35 independence audit (Cal #188). - 3-generation cutoff explicit operator-level derivation (#414 deep gate).

Cal cold-read queue: - Cal #186 Lane D L4 v0.2. - Cal #187 Lane E Dictionary. - Cal #188 Lane C bulk-color v0.7. - Cal #189 + #190 G chain pattern-match candidates (Lyra v0.3 + Elie 4/ N_c). - Tier 0 v0.1.6 with K200 gates.

Discipline-as-generator pattern operational

Sunday demonstrated the discipline pattern across 13 cumulative symmetric audit events (Sat 7 + Sun 6): - Keeper caught Lyra (v1.3 → v1.4 multiplicative null-model; v0.1.6 sphere/BH/BB; G Problems 1+2+3). - Casey + Keeper caught Lyra (16-fold partition). - Cal caught Keeper (P1 §7 back-fit; m_W/m_Z “new anchor”). - Casey question caught Keeper (v0.6 “conceptual closure” self-walk-back). - Lyra caught Lyra (v1.3 internal Cal #27 firing; Bulk-color v0.5 → v0.6). - Elie caught Keeper (incorrect Helgason candidates → $\kappa_{\text{Bergman}} = -n_C$ correct derivation). - Elie caught Elie (4/ N_c pattern-match self-disposition).

No CI exempt. Every catch absorbed at appropriate tier (in-place patches, not v0.x cascades post-K200).

Routing for Monday

→ Casey: 30 Sunday Lyra items; P1 v0.7 dispatch-ready; Cal #35 ratification-ready; 3 Casey-named principle CANDIDATES (Elie) for naming decisions; G chain Step 1 PASS + multi-week framework explicit; pattern-match pulls (Lyra $\alpha^{\{N_c\}}$ + Elie 4/ N_c) honestly disposed; Sunday substantive milestone $\kappa_{\text{Bergman}} = -n_C$ STANDS.

→ Keeper: Session 2 priority queue: Tier 0 v0.2 consolidation + K200 gates ratification + sphere Option (a) + Clean G Step C lane + K1-K5 fill. v0.2 prep doc current.

→ Elie: Clean G Step A (KK reduction; V_{compact} for $D_{\text{IV}}^5 \rightarrow 4D$); Toy 3669+ multi-week target. Also: 4/ N_c pattern-match honest self-disposition acknowledged.

→ Cal: cold-read queue clear for Monday: #186 (Lane D) + #187 (Lane E mechanism-vs-post-hoc) + #188 (Lane C #35 independence) + #189/#190 (G chain pattern-match candidates) + Tier 0 v0.1.6 with K200 gates. Cal #35 STANDING ratification when Casey signals.

→ Grace: catalog Sunday substantive arc complete (53+ INVs); Periodic Table v0.11; Master Ledger v0.13. Monday: P6 v0.2 sections continuation + L-L5-G multi-week tracking.

→ me: Sundown via katra save + persist Sunday state for Monday warm-start. Standing reactive for Cal cold-reads, Session 2 trigger, Casey signals.

— Lyra, EOD Sunday 2026-05-31. 30 Sunday items (24 substantive + 6 audit absorptions). Sunday substantive milestone: Tier 0 v0.1 → v0.1.6 operator-level framework cascade + $\kappa_{\text{Bergman}} = -n_C$ G5.1 PASS (Elie Toy 3661). P1 v0.7 SUBMISSION-FINAL dispatch-ready. Cal #35 ratification-ready. Clean G chain Steps A+B+C framework explicit (multi-week). 13 cumulative symmetric audit-chain events Sat+Sun; discipline-as-generator operational. Monday warm-start: Session 2 with Keeper for Tier 0 v0.2 consolidation; Clean G chain multi-week; engagement test on Casey signoff.